Technical Report: Mercator Internal Knowledge Assistant — Post-Launch Evaluation

Project: MercatorBot v1 (internal codename: "Polica") **Report author:** Žiga Potočnik **Reviewed by:** Maša Vidmar **Date:** 2024-08-19 **Status:** Final — shared with client August 21

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1. Overview

This report covers the 60-day post-launch evaluation of the internal knowledge assistant deployed for Mercator's category management and procurement teams in June 2024. The system was contracted under the Kvant–Mercator MSA signed February 2024, at a total project value of €34,500.

The assistant allows Mercator staff to query a corpus of ~1,100 internal documents: supplier contracts, category performance reports, internal pricing guidelines, and historical promotional campaign data.

Evaluation period: June 17 – August 16, 2024 **Active users during period:** 43 (out of 80 provisioned accounts) **Total queries processed:** 2,847

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2. System Configuration (as deployed)

Component	Choice	Notes
Embedding model	`text-embedding-3-small`	Cost optimization vs. large; acceptable on Slovenian
Vector DB	Qdrant (Hetzner, FSN1)	Single node, 16GB RAM
Chunk size	400 tokens, 80 token overlap	Tuned during dev; original was 512/100
Retrieval	Dense only (no hybrid)	Hybrid was descoped due to timeline pressure
Top-k	6	Increased from 4 after first two weeks
LLM	GPT-4o-mini	GPT-4o was in the original spec — see note below

Component	Choice	Notes
Framework	LlamaIndex 0.10	
Reranker	None	Descoped
Interface	Streamlit (internal only)	
Deployment	Docker, Hetzner CX31	

Note on LLM downgrade: The switch from GPT-4o to GPT-4o-mini was made during week 6 of development at Mercator's request — their IT procurement team objected to the estimated API costs (€600–900/month at projected volumes). In our assessment this was a mistake. Answer quality on complex multi-document queries is noticeably worse with mini. We flagged this in writing (email June 3, 2024) and Mercator accepted the trade-off. Worth revisiting at the next review.

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3. Evaluation Results

3.1 Automated Evaluation (RAGAS)

Evaluation set: 45 questions prepared by Kvant with input from two Mercator category managers. Mix of factual lookups, multi-document synthesis, and unanswerable questions.

Metric	Score	Benchmark
Faithfulness	0.81	>0.80 target
Answer Relevancy	0.74	>0.80 target
Context Precision	0.69	>0.75 target
Context Recall	0.77	>0.75 target
Answer Correctness	0.72	>0.75 target

Faithfulness is acceptable — the system is not making things up badly. The bigger issue is relevancy and precision: it's retrieving chunks but sometimes the retrieved context doesn't match the actual information need. This is a retrieval problem, not a generation problem.

3.2 Query Analysis

Of 2,847 total queries:

Query type (manual sample, n=200)	%	Notes
Simple factual lookup	41%	High success rate
Comparison across suppliers/categories	28%	Mixed results
Historical trend questions	18%	Poor — date handling issues (see §4.2)
Unanswerable / out-of-corpus	13%	Refusal rate: 61% (should be higher)

3.3 User Feedback

Informal feedback collected via Slack channel (#polica-feedback, 34 responses):

- "Works well for quick contract lookups" — general sentiment
- "Gives wrong dates sometimes" — 7 mentions
- "Doesn't know about things from before 2022" — 3 mentions (corpus issue, not model)
- "Sometimes confident when it shouldn't be" — 4 mentions

One category manager (Mateja, dairy/chilled) has stopped using it and gone back to manual search. She was one of the more demanding test users and her queries are mostly multi-hop. Possibly worth a dedicated session with her to understand the failure cases.

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4. Known Issues

4.1 Hybrid Retrieval Gap

The decision to use dense-only retrieval is hurting performance on exact-match queries (e.g., "What is the margin for SKU 4871029?"). Dense retrieval does not reliably surface documents containing specific product codes or numerical identifiers. BM25 or keyword search would solve this. We recommend adding hybrid retrieval in the next iteration.

Estimated effort: 3–5 days of engineering. Cost: ~€2,800 if contracted separately.

4.2 Date Handling

Several documents in the corpus contain relative date references ("last quarter," "this year") that were accurate at time of writing but are now misleading. The system treats these literally and can return incorrect temporal context.

Example: A 2022 promotional campaign report refers to "this year's Christmas campaign." A query about the "2024 Christmas campaign" has a non-trivial chance of surfacing this document and generating a confused or incorrect answer.

Mitigation: Add document date as mandatory metadata, include in prompt context. Also flag documents older than 18 months with a warning in the UI. Both are low-effort changes.

4.3 Refusal Rate on Unanswerable Questions

The system refuses to answer approximately 61% of questions that are genuinely out of scope (i.e., the information is not in the corpus). This means it hallucinates or confabulates an answer ~39% of the time on unanswerable questions. This is too high.

Root cause: The retrieval confidence threshold is set too low (cosine similarity > 0.72). Increasing to 0.78 in testing reduced hallucination rate to ~18% on the unanswerable set, with a small increase in false refusals on answerable questions. Recommend threshold increase.

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5. Recommendations

Priority	Recommendation	Effort	Est. Cost
High	Add BM25 hybrid retrieval	4 days	€2,400
High	Increase retrieval confidence threshold to 0.78	0.5 days	€300
High	Add document date metadata to prompt context	1 day	€600
Medium	Add "document may be outdated" warning for docs >18 months	1 day	€600
Medium	Upgrade LLM to GPT-4o for complex query path	—	+~€400/month API
Low	Add reranker (BGE)	3 days	€1,800

Priority	Recommendation	Effort	Est. Cost
Low	Dedicated session with Mateja (dairy) to understand failure cases	0.5 days	€300

Total estimated improvement cost: **€5,700 + ongoing API delta**

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6. Conclusion

The system is functioning and is being used, which is the baseline. However, it is underperforming on the metrics that matter most for complex queries. The issues are diagnosed and fixable. None of them require architectural changes — they are tuning, configuration, and retrieval improvements.

The "works fine for simple lookups" user sentiment suggests the system has established a foothold. The risk is that power users (like Mateja) will abandon it before improvements are made, which will affect internal perception.

Recommend scheduling a follow-up call with Mercator by September 5 to present this report and agree on next steps.

Attachments: RAGAS evaluation script (GitHub), evaluation Q&A set (Google Sheet, shared with Mercator PM), query log sample (anonymized, 200 queries)